

The Correlation Coefficient and Independence

- Let X and Y have joint pdf $f(x, y)$.
- μ_1 and μ_2 are the means of X and Y , σ_1^2 and σ_2^2 are the variances of X and Y .
- $u(x, y)$ is a function of x and y .

- Covariance:

$$\begin{aligned}\text{cov}(X, Y) &= \mathbb{E}[(X - \mu_1)(Y - \mu_2)] = \mathbb{E}(XY - \mu_2 X - \mu_1 Y + \mu_1 \mu_2) \\ &= \mathbb{E}(XY) - \mu_2 \mathbb{E}(X) - \mu_1 \mathbb{E}(Y) + \mu_1 \mu_2 = \mathbb{E}(XY) - \mu_1 \mu_2.\end{aligned}$$

- Correlation Coefficient:

$$\rho = \frac{\mathbb{E}[(X - \mu_1)(Y - \mu_2)]}{\sigma_1 \sigma_2} = \frac{\text{cov}(X, Y)}{\sigma_1 \sigma_2}.$$

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$$\mathbb{E}(XY) = \mu_1 \mu_2 + \rho \sigma_1 \sigma_2 = \mu_1 \mu_2 + \text{cov}(X, Y).$$

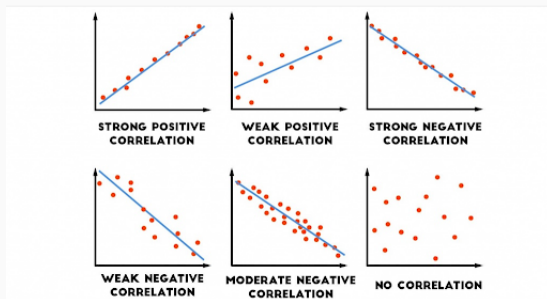
- Let X and Y have the joint pdf

$$f(x, y) = \begin{cases} x + y & 0 < x < 1, 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Find the correlation coefficient ρ of X and Y .

$\mu_1 = \mathbb{E}(X) = \int_0^1 \int_0^1 x(x + y) dx dy = \frac{7}{12}$ and
 $\sigma_1^2 = \mathbb{E}(X^2) - \mu_1^2 = \int_0^1 \int_0^1 x^2(x + y) dx dy - (\frac{7}{12})^2 = \frac{11}{144}$. Similarly, $\mu_2 = \frac{7}{12}$
and $\sigma_2^2 = \frac{11}{144}$. The covariance of X and Y is
 $\mathbb{E}(XY) - \mu_1\mu_2 = \int_0^1 \int_0^1 xy(x + y) dx dy - (\frac{7}{12})^2 = -\frac{1}{144}$. The correlation
coefficient of X and Y is

$$\rho = \frac{-\frac{1}{144}}{\sqrt{(\frac{11}{144})(\frac{11}{144})}} = -\frac{1}{11}$$



- The correlation coefficient ρ proves to be a very useful characteristic of the relation between X and Y .
- $-1 \leq \rho \leq 1$.
- If $\rho = 1$, $\mathbb{P}(Y = a + bX) = 1$, $b > 0$.
- If $\rho = -1$, $\mathbb{P}(Y = a + bX) = 1$, $-b > 0$.
- ρ is a measure of the intensity of the concentration of the probability for X and Y about a line.

- Let X_1 and X_2 denote the random variables of continuous type which has joint pdf $f(x_1, x_2)$ and marginal pdf $f_1(x_1)$ and $f_2(x_2)$, respectively.
- Conditional pdf $f_{2|1}(x_2|x_1)$: $f(x_1, x_2) = f_{2|1}(x_2|x_1)f_1(x_1)$.
- Suppose that $f_{2|1}(x_2|x_1)$ does **NOT** depend on x_1 , then

$$f_2(x_2) = \int f_{2|1}(x_2|x_1)f_1(x_1)dx_1 = f_{2|1}(x_2|x_1) \int f_1(x_1)dx_1 = f_{2|1}(x_2|x_1)$$

$$f_2(x_2) = f_{2|1}(x_2|x_1); \quad f(x_1, x_2) = f_1(x_1)f_2(x_2)$$

- The same discussion applies to the discrete case too.
- **Independence:** X_1 and X_2 are said to be independent if and only if $f(x_1, x_2) = f_1(x_1)f_2(x_2)$ or $p(x_1, x_2) = p_1(x_1)p_2(x_2)$. Random variables that are not independent are said to be **dependent**.

- Let X_1 and X_2 have the joint pdf

$$f(x_1, x_2) = \begin{cases} x_1 + x_2 & 0 < x_1 < 1, 0 < x_2 < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Check if X_1 and X_2 are independent or dependent?

The marginal pdf of X_1 is $f_1(x_1) = \int_0^1 (x_1 + x_2) dx_2 = x_1 + \frac{1}{2}$, $0 < x_1 < 1$.

Similarly, the marginal pdf of X_2 is $f_2(x_2) = \int_0^1 (x_1 + x_2) dx_1 = x_2 + \frac{1}{2}$, $0 < x_2 < 1$. So, $f(x_1, x_2) \neq f_1(x_1)f_2(x_2)$, and they are not independent.

- Let X and Y have the joint pdf $f(x, y) = 4xy$, $0 < x < 1$, $0 < y < 1$ zero elsewhere. Are X and Y independent?

The marginal distribution of X is $f_1(x) = 2x$, $0 < x < 1$. The marginal distribution of Y is $f_2(y) = 2y$, $0 < y < 1$. Therefore $f(x, y) = f_1(x)f_2(y)$ for any (x, y) . They are independent.

- Let X_1 and X_2 have the support $\mathcal{S}_1$ and $\mathcal{S}_2$, respectively, and have the joint pdf $f(x_1, x_2)$. Then X_1 and X_2 are independent if and only if $f(x_1, x_2)$ can be written as a product of a nonnegative function of x_1 and a nonnegative function of x_2 ,

$$f(x_1, x_2) = g(x_1)h(x_2),$$

where $g(x_1) > 0$, $x_1 \in \mathcal{S}_1$ and $h(x_2) > 0$, $x_2 \in \mathcal{S}_2$, zero elsewhere.

- This theorem is true for the discrete case also.

- Check previous examples. $f(x_1, x_2) = x_1 + x_2$, $0 < x_1, x_2 < 1$;
 $f(x, y) = 4xy$, $0 < x, y < 1$.
- Let the joint pdf be $f(x_1, x_2) = 8x_1x_2$, $0 < x_1 < x_2 < 1$, zero elsewhere.
The formula $8x_1x_2$ might suggest to some that X_1 and X_2 are independent. However, if we consider the space $\mathcal{S} = \{0 < x_1 < x_2 < 1\}$, we see it is not a product space. This should make it clear that, in general, X_1 and X_2 must be dependent if the space of positive probability density of X_1 and X_2 is bounded by a curve that is neither a horizontal nor vertical line.

- Suppose X_1 and X_2 are independent and that $\mathbb{E}(u(X_1))$ and $\mathbb{E}(v(X_2))$ exist. Then,

$$\mathbb{E}[u(X_1)v(X_2)] = \mathbb{E}[u(X_1)]\mathbb{E}[v(X_2)].$$

- Attention: The conclusion is only true for one direction.
- If X and Y are independent, then the correlation coefficient or covariance of X and Y are zero. If the covariance is zero, are they independent?
- A counterexample: $p(x, y) = \frac{1}{3}$, $(x, y) = (0, 0), (1, 1), (2, 0)$, zero elsewhere.

- Suppose the joint mgf, $M(t_1, t_2)$ exists for the random variables X_1 and X_2 . Then X_1 and X_2 are independent if and only if

$$M(t_1, t_2) = M(t_1, 0)M(0, t_2).$$

- Let (X, Y) be a pair of random variable with joint pdf

$$f(x, y) = \begin{cases} e^{-x} & 0 < x < y < \infty \\ 0 & \text{elsewhere.} \end{cases}$$

$$M(t_1, t_2) = \frac{1}{(1 - t_1 - t_2)(1 - t_2)},$$

where $t_1 + t_2 < 1$ and $t_2 < 1$. Because $M(t_1, t_2) \neq M(t_1, 0)M(0, t_2)$ the random variables are dependent.

References

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- [3] *A First Course in Probability* by Ross
- [4] *John E. Freund's Mathematical Statistics with Applications* by Irwin Miller Marylees Miller